Bayesian Media Mix Model for Company X

Data Science Challenge | Cagan Yigit Deliktas | October 2026

Key results

- Carry-over is short: 19 to 26% of a channel's effect comes after the week of the spend.
- Paid media drives about **39% of revenue**. The rest is baseline (trend, season, Cyber Week).
- On 16 unseen weeks the model has **11.5% error**, vs 24.7% for a seasonal naive forecast.
- **ch5 has the best ROI** (ROAS ≈ 1.22). ch6 is about break-even, ch4 and ch3 return the least.

Model

$$\text{revenue}_t = \beta_0 + \beta_{trend}\, t + \text{season}_t + \beta_{cw}\, \text{CyberWeek}_t + \sum_{c=1}^{7} \text{ROAS}_c \cdot \text{adstock}_{c,t} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma)$$

- **Data:** 104 weeks (Aug 2020 to Aug 2022), 7 channels. Total spend is 51% of revenue.
- **Scaling:** revenue and spend are divided by the same number (average weekly revenue, 136,490 €). So each channel coefficient is directly the ROAS (€ revenue per € spend).
- **Baseline:** linear trend, yearly seasonality (2 Fourier pairs) and a Cyber Week indicator. Cyber Week is 2 to 3 times a normal week and spend also jumps then, so without this term the ads would get the credit.
- **Sampling:** NUTS, 4 chains × 1000 draws after 1000 tuning steps, target_accept = 0.95, seed 42.

1. How do you model spend carry-over?

With a **geometric adstock**: one decay rate α per channel and an 8-week window.

$$\text{adstock}_{c,t} = \sum_{l=0}^{7} w_{c,l}\, \text{spend}_{c,t-l}, \qquad w_{c,l} = \frac{\alpha_c^l}{\sum_{k=0}^{7} \alpha_c^k}$$

- Each week keeps a share α of the previous week's effect. It is the standard MMM choice and needs only one parameter per channel.
- α is per channel, because channels can act at different times.
- The weights sum to 1. Adstock only moves spend over time, so the coefficient stays ROAS.
- No saturation, as the README suggests. Spend before the first week is set to 0.

Result: the posterior means of α are 0.19 to 0.26, so most of the effect lands in the spend week (for α = 0.25 the weights are 0.75, 0.19, 0.05, ...). But the 94% intervals are wide (about 0.01 to 0.7) and close to the prior. The data supports *short* carry-over, but it cannot rank channels by delay: spend is partly correlated (up to 0.65 for ch5 and ch6) and timing is harder to learn from 104 weeks than size.

2. Explain your choice of prior inputs to the model

All priors are **weakly informative** and set on the scaled data, where 1.0 = an average week (136,490 €). Each scale comes from a simple fact about the data: the prior is centred where I have no strong belief and is wide enough to contain what the data shows.

Parameter	Prior	Why these numbers
α (decay)	Beta(1, 3)	Beta lives on [0, 1], like a decay rate. Mean 0.25 means 75% of the effect lands in the spend week. Density is highest at 0, so "no carry-over" stays plausible, and P(α > 0.5) = 12.5%, so long carry-over is possible if the data supports it.
ROAS	HalfNormal(1)	Ads cannot lower revenue, so ROAS ≥ 0. Spend is 51% of revenue, so even if ads made *all* revenue the average ROAS would be 1 / 0.51 ≈ 2. Scale 1 puts 95% of the mass below 2. The prior mean 0.8 implies a media share of about 40%, a neutral start.
Intercept	Normal(0.5, 0.5)	Baseline level in average weeks. 0.5 = about half of revenue without ads, in line with the 40% media share above. sd 0.5 keeps both 0 (no baseline) and 1 (all revenue is baseline) within one sd.

Parameter	Prior	Why these numbers
Trend	Normal(0, 0.5)	t goes from 0 to 1, so the weight is the total change over 2 years. Centred at 0: no direction assumed. Revenue falls from about 1.3 (first 8 weeks) to 0.6 (last 8), and a change of this size is within 2 sd (±1).
Season (4)	Normal(0, 0.3)	Each weight multiplies a sin or cos wave between −1 and 1, so it is the swing in average weeks. Monthly mean revenue goes from 0.64 (July) to 1.22 (November), about ±0.3 around 1. With sd 0.3 a sin/cos pair has a median swing of 0.35 and 95% below 0.73: wide enough for this, but no absurd seasons.
Cyber Week	Normal(0, 1)	Extra revenue in that week, in average weeks. The two Cyber Weeks are 3.1 and 2.0 vs about 1.0 to 1.2 around them, so the extra is 1 to 2 weeks. sd 1 covers this within 2 sd. A narrow prior would push the rest of the spike onto the ads.
σ (noise)	HalfNormal(0.3)	Must be positive. The total sd of weekly revenue is 0.37, so noise cannot be much larger. Median 0.20 and 95% below 0.59: room for noisy weeks, but not more than all the variation.

Check: all posteriors landed well inside these ranges (season −0.12 to +0.08, trend −0.27, Cyber Week 1.23, σ 0.24), so no prior was forcing the result. The prior predictive check (Question 3) confirms the priors give realistic revenue.

3. How are your model results based on prior sampling vs. posterior sampling?

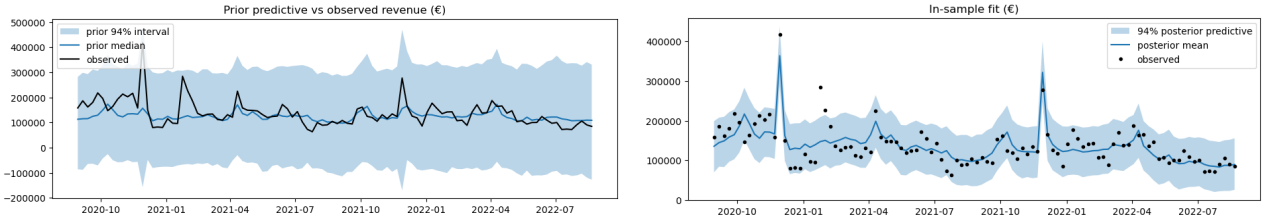


Figure 1. Left: prior predictive (priors only). Right: posterior predictive. Black: observed revenue.

- **Prior predictive:** a wide band that covers the data, with a realistic median. The implied media share is about 40%, so the priors are neutral. It allows some negative revenue (a weakness of the Normal likelihood).
- **Posterior predictive:** tight around the data, no negative values, 97% of weeks inside the 94% band.

For the parameters, the sd ratio shows how much the data narrowed the prior (1 = nothing learned).

	ch1	ch2	ch3	ch4	ch5	ch6	ch7
ROAS prior mean	0.77	0.77	0.82	0.77	0.78	0.80	0.78
ROAS posterior mean	0.55	0.81	0.64	0.47	1.22	0.97	0.76
ROAS sd ratio (post / prior)	0.77	1.03	0.65	0.65	1.05	0.91	0.66
α sd ratio (post / prior)	1.06	0.98	0.92	0.97	0.76	0.94	0.93

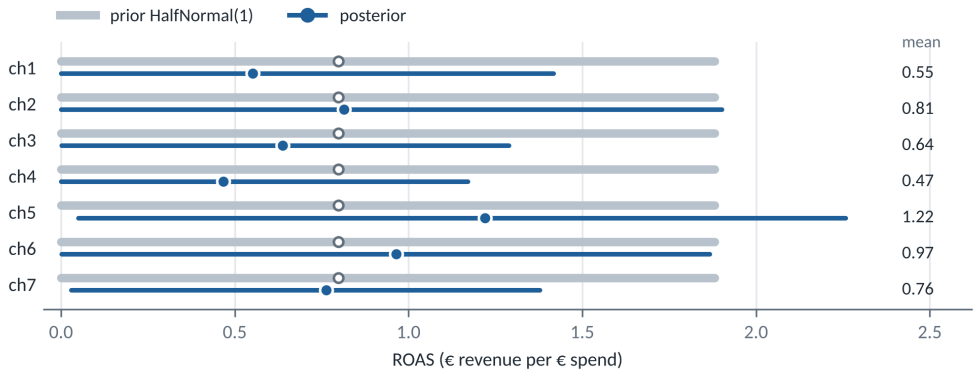


Figure 2. ROAS per channel: prior vs posterior (mean and 94% HDI).

- **Baseline is learned strongly:** Cyber Week adds about 167k € in that week, and the trend lowers weekly revenue by about 36k € over two years.

- **ch3, ch4, ch7** (large budgets): about 35% narrower, ch3 and ch4 move down.
- **ch5** moves up (0.78 → 1.22) but does not get narrower.
- **ch1, ch2** (1.8% and 0.5% of budget) and **all α**: close to the prior. The data says little here.

4. How good is your model performing? How do you measure it?

- **Sampler:** 0 divergences, R-hat = 1.00 and ESS above 1000 for all parameters, so the samples are reliable.
- **In-sample:** R^2 = 0.64, MAPE = 17.4%, and 97% of weeks are inside the 94% band (well calibrated, slightly wide).
- **Out-of-sample:** fit on the first 88 weeks, predict the last 16 (May to Aug 2022). A time split, so the model cannot see the future.

Forecast of the last 16 weeks	MAPE	94% coverage
Full model	11.5%	100%
Model without the 7 channels	13.5%	100%
Seasonal naive (same week last year)	24.7%	–
Naive (mean of last 16 training weeks)	52.4%	–

The model has half the error of the seasonal naive forecast. Without the channels the error is 13.5%, so most predictive power comes from trend, season and Cyber Week; the channels add a modest 2 points.

Where it misses: a 3-week jump in January 2021 with flat spend (no driver in the data); the shared Cyber Week effect (too low for 2020: 365k vs 418k €, too high for 2021: 323k vs 277k €); and the first weeks, which have no spend history.

5. What are your main insights in terms of channel performance / effects?

- **Paid media drives 39% of revenue** (94% interval 24 to 55%). The baseline has a falling trend, a summer low and peaks in spring and autumn. Spend also peaks in spring and autumn, so the seasonal terms stop the ads from taking that credit.
- **ch5 and ch6** bring more than their budget share, **ch3 and ch4** less, **ch7** is at par. ch1 and ch2 are too small to judge.

	ch1	ch2	ch3	ch4	ch5	ch6	ch7
Share of spend	1.8%	0.5%	28.1%	10.0%	12.4%	7.3%	39.9%
Share of media revenue	1.4%	0.6%	23.4%	6.2%	20.1%	9.7%	38.7%

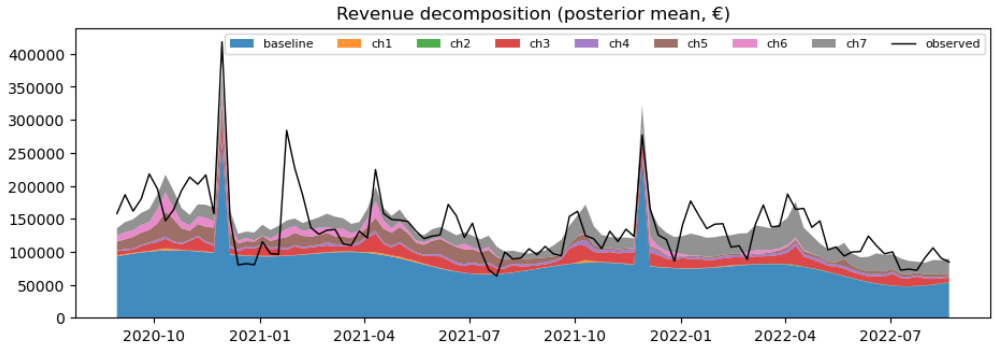


Figure 3. Weekly revenue split into baseline and channel contributions (posterior mean).

Above average is not the same as profitable: the average media ROAS is only about 0.76.

6. Can you derive ROI estimates per channel? What is the best channel in terms of ROI?

Yes. For each of the 4000 posterior samples: ROAS = channel contribution summed over all weeks / channel spend, and ROI = ROAS − 1. This gives a full distribution per channel. P(ROAS > 1) is the chance that a channel pays back its spend; P(best) is the share of samples where it has the highest ROAS.

Channel	Spend (€)	ROAS	94% interval	ROI	P(ROAS > 1)	P(best)
ch5	891,864	1.22	0.18 – 2.43	+22%	0.61	0.41
ch6	526,625	0.97	0.10 – 2.09	−3%	0.44	0.22
ch2	35,739	0.81	0.04 – 2.17	−19%	0.33	0.14
ch7	2,880,942	0.76	0.11 – 1.50	−24%	0.26	0.08
ch3	2,028,747	0.64	0.05 – 1.42	−36%	0.18	0.05
ch1	129,543	0.55	0.02 – 1.66	−45%	0.17	0.06
ch4	719,174	0.47	0.02 – 1.40	−53%	0.10	0.03

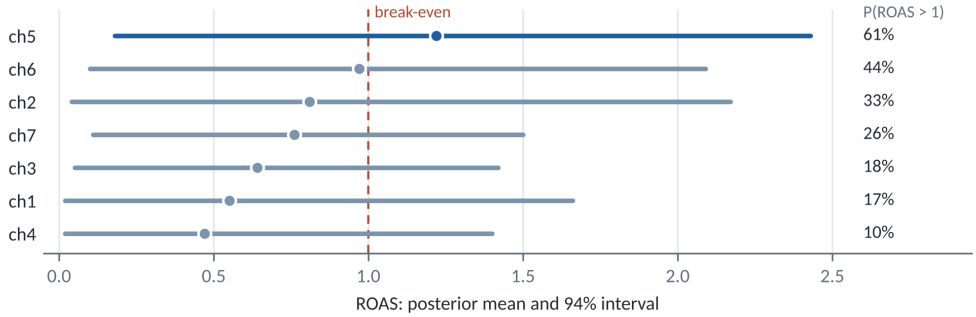


Figure 4. Posterior ROAS per channel, best first.

ch5 is the best channel: ROAS ≈ 1.22 (ROI +22%), 61% chance to pay back its spend, and the best channel in 41% of samples. But the intervals overlap a lot, so the ranking is a probability, not a certainty.

- The ROI is **revenue-based**. With a margin m, a channel breaks even only if m × ROAS > 1 (with 30% margin: ROAS > 3.3).
- Without saturation these are **average** returns, not the return of the next €.

Limitations and how to reduce them

Limitation	What could help
Observational data: missing drivers (promotions, price, competitors) can bias the estimates, as January 2021 shows.	Add these drivers as covariates, and calibrate the ROAS priors with lift tests (e.g. pausing a channel in some regions).
Correlated spend (up to 0.65) makes some channels hard to separate, and α is barely learned.	Informative priors from experiments, or planned budget changes so that channels move independently.
Normal likelihood allows negative revenue and assumes the same noise every week.	Model log revenue (log-normal) or use a Student-t likelihood, which is robust to outlier weeks.
No saturation, so ROAS is an average return, not the return of the next €.	Add a saturation curve per channel (e.g. Hill or logistic) to get marginal ROAS for budget decisions.
One shared Cyber Week effect: too low for 2020, too high for 2021.	A separate effect per year, or a promotion/holiday indicator with its intensity.
Only 2 years of data and no spend history before the first week.	Use more history, or leave the first 7 weeks out of the likelihood as an adstock warm-up.